Nama : Tsaniya Faridah Maulida

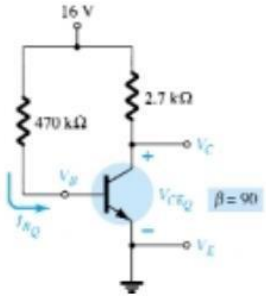
NIM : 211331031

Kelas : 1A-TTE

Soal BAB 4 (buku electronic devices and circuit theory, R boylestad) 1 – 32

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1. **For he fixed-bias configuration of Fig. 4.73, determine:**
   a. IBQ
   b. ICQ
   c. VCEQ
   d. VC
   e. VB
   f. VE



   Jawab :

   a) IBQ = $\frac{Vcc - VBE}{RB}$ = = $\frac{16\ V - 0{,}7\ V}{510 k\Omega}$ = $\frac{15{,}3\ V}{510\ k\Omega}$ = 30 μA

   b) ICQ = βIBQ = (120)(30 μA) = 3,6 mA

   c) VCEQ = Vcc – ICQ Rc = 16 V – (3,6 mA)(1,8 kΩ) = 9,52 V

   d) VC = VCEQ = 6,48 V

   e) VB = VBE = 0,7 V

   f) VE = 0 V

2. **Given the information appearing in Fig. 4.74, determine:**
   a. IC
   b. Rc
   c. RB
   d. VCE
   Jawab :

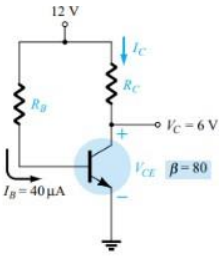


Figure 4.74 Problem 2

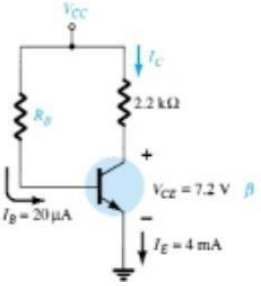
   a) IC = βIB = 80(40 μA) = 3,2 mA

   b) Rc = $\frac{VRC}{Ic}$ = = $\frac{Vcc - Vc}{Ic}$ = $\frac{12V - 6V}{3{,}2\ mA}$ = $\frac{6V}{3{,}2\ mA}$ = 1.875 kΩ

   c) RB = $\frac{VRB}{IB}$ = $\frac{12\ V - 0{,}7\ V}{40\ \mu A}$ = $\frac{11{,}3\ V}{40\ \mu A}$ = 282.5 kΩ

   d) VCE = VCE = 6 V

3. **Given the information appearing in Fig. 4.75, determine:**
   a. IC
   b. Vcc
   c. β
   d. RB
   Jawab :



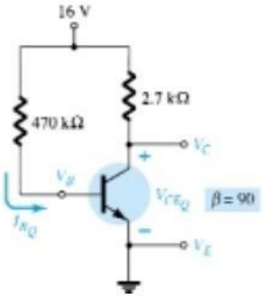
   a) IC = IE – IB = 4 mA – 20 μA = 3.98 mA ≅ 4 mA

   b) VCC = VCE + ICRC = 7,2 V + (3,98 mA)(2,2 kΩ)

= 15.96 V ≅ 16 V

c) $\beta = \frac{IC}{IB} = \frac{3{,}98\ mA}{20\ \mu A} = 199 \cong 200$

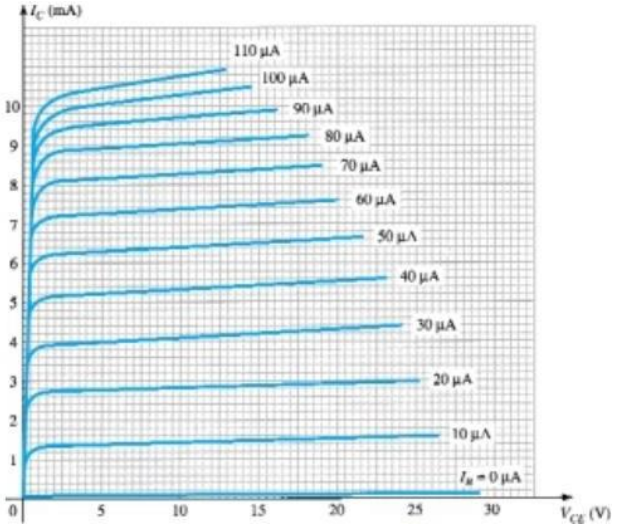
d) RB = $\frac{VRB}{IB} = \frac{Vcc-VBE}{IB} = \frac{15{,}96\ V-0{,}7\ V}{20\ \mu A} = 763\ k\Omega$



4. **Find the saturation current (ICsat) for the fixed-bias configuration of Fig. 4.73.**

   Jawab : ICsat = $\frac{Vcc}{RC} = \frac{16\ V}{2{,}7\ k\Omega} = 5.93$ mA

5. **Given the BJT transistor characteristics of Fig. 4.76:**
   a. Draw a load line on the characteristics determined by E = 21 V and RC = 3 k for a fixed-bias configuration.
   b. Choose an operating point midway between cutoff and saturation. Determine the value of RB to establish the resulting operating point.
   c. What are the resulting values of ICQ and VCEQ?
   d. What is the value of β at the operating point?
   e. What is the value of α defined by the operating point?
   F. What is the saturation (ICsat) current for the design?
   G. Sketch the resulting fixed-bias configuration.
   H. What is the dc power dissipated by the device at the operating point?
   I. What is the power supplied by VCC?
   J. Determine the power dissipated by the resistive elements by taking the difference between the results of parts (h) and (i).



Jawab :

a) Load line intersects vertical axis at IC = $\frac{21\ V}{3\ k\Omega}$ = 7 mA and horizontal axis at

VCE = 21 V.

b) IB = 25 μA: RB = $\frac{Vcc-VBE}{IB}$ = = $\frac{21\ V-0,7\ V}{25\ \mu A}$ = 812 kΩ

c) ICQ ≅ 3,4 mA, VCEQ ≅ 10,75 V

d) $\beta = \frac{IC}{IB} = \frac{3,4\ mA}{25\ \mu A}$ = 136

e) $\alpha = \frac{\beta}{\beta+1} = \frac{136}{136+1} = \frac{136}{137}$ = 0,992

f) ICsat = $\frac{Vcc}{RC} = \frac{21\ V}{3\ k\Omega}$ = 7 mA

g) –

h) PD = VCEQ . ICQ = (10,75 V)(3,4 mA) = 36,55 mW

i) Ps = Vcc (Ic + IB) = 21 V(3,4 mA + 25 μA) = 71,92 mW

j) PR = Ps – PD = 71,92 mW – 36,55 mW = 35,37 mW

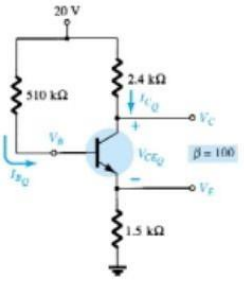


Figure 4.77 Problems 6, 9, 11, 20, 24, 48, 51, 54

6. For the emitter-stabilized bias circuit of Fig. 4.77, determine:

a. IBQ
b. ICQ
c. VCEQ
d. Vc
e. VB
f. VE

Jawab :

a) IBQ = $\frac{Vcc-VBE}{RB+(\beta+1)RE} = \frac{20\ V-0,7}{510\ k\Omega+(101)1,5\ k\Omega}$

= $\frac{19,3\ V}{661,5\ k\Omega}$ = 29,18 μA

b) ICQ = βIBQ = (100)(29,18 μA) = 2,92 mA

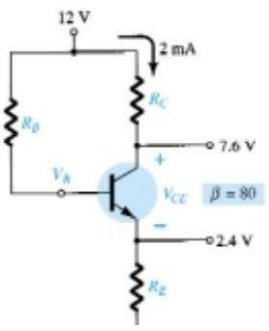
c) VCEQ = VCC − IC(RC + RE) = 20 V − (2,92 mA)(2,4 kΩ + 1,5 kΩ)
= 20 V – 11,388 V
= 8,61 V

d) VC = VCC – IC.RC = 20 V − (2,92 mA)(2,4 kΩ) = 20 V – 7,008 V = 13 V

e) VB = VCC – IB.RB = 20 V − (29,18 μA)(510 kΩ)
= 20 V – 14,882 V = 5,12 V

f) VE = VC − VCE = 13 V – 8,61 V = 4,39 V

7. Given the information provided in Fig. 4.78, determine:



(Mengingat informasi yang diberikan pada Gambar 4.78, tentukan)

a. RC
b. RE
c. RB
d. VCE
e. VB.

Jawab:

(a) $R_C = \dfrac{V_{CC} - V_C}{I_C} = \dfrac{12\text{ V} - 7.6\text{ V}}{2\text{ mA}} = \dfrac{4.4\text{ V}}{2\text{ mA}} = \mathbf{2.2\ k\Omega}$

(b) $I_E \cong I_C$: $R_E = \dfrac{V_E}{I_E} = \dfrac{2.4\text{ V}}{2\text{ mA}} = \mathbf{1.2\ k\Omega}$

(c) $R_B = \dfrac{V_{R_B}}{I_B} = \dfrac{V_{CC} - V_{BE} - V_E}{I_B} = \dfrac{12\text{ V} - 0.7\text{ V} - 2.4\text{ V}}{2\text{ mA}/80} = \dfrac{8.9\text{ V}}{25\ \mu\text{A}} = \mathbf{356\ k\Omega}$

(d) $V_{CE} = V_C - V_E = 7.6\text{ V} - 2.4\text{ V} = \mathbf{5.2\ V}$

(e) $V_B = V_{BE} + V_E = 0.7\text{ V} + 2.4\text{ V} = \mathbf{3.1\ V}$

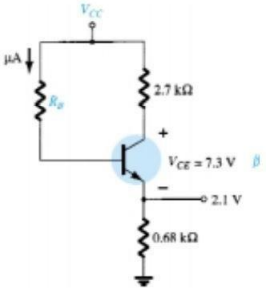


Figure 4.79 Problem 8

8. Given the information provided in Fig. 4.79, determine:

a. β
b. VCC
c. RB

Jawab:

(a) $I_C \cong I_E = \dfrac{V_E}{R_E} = \dfrac{2.1\text{ V}}{0.68\text{ k}\Omega} = 3.09\text{ mA}$

$\beta = \dfrac{I_C}{I_B} = \dfrac{3.09\text{ mA}}{20\ \mu\text{A}} = \mathbf{154.5}$

(b) $V_{CC} = V_{R_C} + V_{CE} + V_E$

$= (3.09\text{ mA})(2.7\text{ k}\Omega) + 7.3\text{ V} + 2.1\text{ V} = 8.34\text{ V} + 7.3\text{ V} + 2.1\text{ V}$

$= \mathbf{17.74\ V}$

(c) $R_B = \dfrac{V_{R_B}}{I_B} = \dfrac{V_{CC} - V_{BE} - V_E}{I_B} = \dfrac{17.74\text{ V} - 0.7\text{ V} - 2.1\text{ V}}{20\ \mu\text{A}}$

$= \dfrac{14.94\text{ V}}{20\ \mu\text{A}} = \mathbf{747\ k\Omega}$

9. Determine the saturation current (ICsat) for the network of Fig. 4.77.

Jawab:

$$I_{C_{sat}} = \frac{V_{CC}}{R_C + R_E} = \frac{20\text{ V}}{2.4\text{ k}\Omega + 1.5\text{ k}\Omega} = \frac{20\text{ V}}{3.9\text{ k}\Omega} = \mathbf{5.13\ mA}$$

10. Using the characteristics of Fig. 4.76, determine the following for an emitter-bias configuration if a Q-point is defined at ICQ 4 mA and VCEQ 10 V.

a. RC if VCC 24 V and RE 1.2 k .
b. β at the operating point.
c. Rn.
d. Power dissipated by the transistor.
e. Power dissipated by the resistor RC.

Jawab:

(a) $I_{C_{sat}} = 6.8\text{ mA} = \dfrac{V_{CC}}{R_C + R_E} = \dfrac{24\text{ V}}{R_C + 1.2\text{ k}\Omega}$

$$R_C + 1.2\text{ k}\Omega = \frac{24\text{ V}}{6.8\text{ mA}} = 3.529\text{ k}\Omega$$

$$R_C = \mathbf{2.33\ k\Omega}$$

(b) $\beta = \dfrac{I_C}{I_B} = \dfrac{4\text{ mA}}{30\ \mu\text{A}} = \mathbf{133.33}$

(c) $R_B = \dfrac{V_{R_B}}{I_B} = \dfrac{V_{CC} - V_{BE} - V_E}{I_B} = \dfrac{24\text{ V} - 0.7\text{ V} - (4\text{ mA})(1.2\text{ k}\Omega)}{30\ \mu\text{A}}$

$$= \frac{18.5\text{ V}}{30\ \mu\text{A}} = \mathbf{616.67\ k\Omega}$$

(d) $P_D = V_{CE_Q} I_{C_Q}$

$$= (10\text{ V})(4\text{ mA}) = \mathbf{40\ mW}$$

(e) $P = I_C^2 R_C = (4\text{ mA})^2(2.33\text{ k}\Omega)$

$$= \mathbf{37.28\ mW}$$

11. (a) Determine IC and VCE for the network of Fig. 4.73.

(b) Change ———————————— to 135 and determine the new value of IC and VCE for the network of Fig. 4.73.

(c) Determine the magnitude of the percent change in IC and VCE using the following equations

$$\%\Delta I_C = \left|\frac{I_{C(\text{part b})} - I_{C(\text{part a})}}{I_{C(\text{part a})}}\right| \times 100\%, \qquad \%\Delta V_{CE} = \left|\frac{V_{CE(\text{part b})} - V_{CE(\text{part a})}}{V_{CE(\text{part a})}}\right| \times 100\%$$

(d) Determine IC and VCE for the network of Fig. 4.77.

(e) Change ———————————— to 150 and determine the new value of IC and VCE for the network of Fig. 4.77.

(f) Determine the magnitude of the percent change in IC and VCE using the following equations:

$$\%\Delta I_C = \left|\frac{I_{C(\text{part e})} - I_{C(\text{part d})}}{I_{C(\text{part d})}}\right| \times 100\%, \qquad \%\Delta V_{CE} = \left|\frac{V_{CE(\text{part e})} - V_{CE(\text{part d})}}{V_{CE(\text{part d})}}\right| \times 100\%$$

(g) In each of the above, the magnitude of ———————————— was increased 50%. Compare the percent change in IC and VCE for each configuration, and comment on which seems to be less sensitive to changes in β

Jawab:

(a) Problem 1: $I_{C_Q} = \mathbf{2.93\ mA}$, $V_{CE_Q} = \mathbf{8.09\ V}$

(b) $I_{B_Q} = 32.55\ \mu\text{A}$ (the same)

$$I_{C_Q} = \beta I_{B_Q} = (135)(32.55\ \mu\text{A}) = 4.39\ \text{mA}$$

$$V_{CE_Q} = V_{CC} - I_{C_Q}R_C = 16\ \text{V} - (4.39\ \text{mA})(2.7\ \text{k}\Omega) = \mathbf{4.15\ V}$$

(c) $$\%\Delta I_C = \left|\frac{4.39\ \text{mA} - 2.93\ \text{mA}}{2.93\ \text{mA}}\right| \times 100\% = \mathbf{49.83\%}$$

$$\%\Delta V_{CE} = \left|\frac{4.15\ \text{V} - 8.09\ \text{V}}{8.09\ \text{V}}\right| \times 100\% = \mathbf{48.70\%}$$

Less than 50% due to level of accuracy carried through calculations.

(d) Problem 6: $I_{C_Q} = \mathbf{2.92\ mA}$, $V_{CE_Q} = \mathbf{8.61\ V}$ ($I_{B_Q} = 29.18\ \mu\text{A}$)

(e) $$I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20\ \text{V} - 0.7\ \text{V}}{510\ \text{k}\Omega + (150 + 1)(1.5\ \text{k}\Omega)} = 26.21\ \mu\text{A}$$

$$I_{C_Q} = \beta I_{B_Q} = (150)(26.21\ \mu\text{A}) = \mathbf{3.93\ mA}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E)$$

$$= 20\ \text{V} - (3.93\ \text{mA})(2.4\ \text{k}\Omega + 1.5\ \text{k}\Omega) = \mathbf{4.67\ V}$$

(f) $$\%\Delta I_C = \left|\frac{3.93\ \text{mA} - 2.92\ \text{mA}}{2.92\ \text{mA}}\right| \times 100\% = \mathbf{34.59\%}$$

$$\%\Delta V_{CE} = \left|\frac{4.67\ \text{V} - 8.61\ \text{V}}{8.61\ \text{V}}\right| \times 100\% = \mathbf{46.76\%}$$

(g) For both I_C and V_{CE} the % change is less for the emitter-stabilized.

12. For the voltage-divider bias configuration of Fig. 4.80, determine:

(a) IBQ.

(b) ICQ.

(c) VCEQ.

(d) VC.

(e) VE.

(f) VB.

Jawab:

$\beta R_E \geq 10R_2$
$(80)(0.68\ \text{k}\Omega) \geq 10(9.1\ \text{k}\Omega)$
$54.4\ \text{k}\Omega \not\geq 91\ \text{k}\Omega$ (No!)

(a) Use exact approach:

$$R_{Th} = R_1 \| R_2 = 62\ \text{k}\Omega \| 9.1\ \text{k}\Omega = 7.94\ \text{k}\Omega$$

$$E_{Th} = \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{(9.1\ \text{k}\Omega)(16\ \text{V})}{9.1\ \text{k}\Omega + 62\ \text{k}\Omega} = 2.05\ \text{V}$$

$$I_{B_Q} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05\ \text{V} - 0.7\ \text{V}}{7.94\ \text{k}\Omega + (81)(0.68\ \text{k}\Omega)}$$

$$= \mathbf{21.42\ \mu A}$$

(b) $I_{C_Q} = \beta I_{B_Q} = (80)(21.42\ \mu\text{A}) = \mathbf{1.71\ mA}$

(c) $V_{CE_Q} = V_{CC} - I_{C_Q}(R_C + R_E)$
$= 16\ \text{V} - (1.71\ \text{mA})(3.9\ \text{k}\Omega + 0.68\ \text{k}\Omega)$
$= \mathbf{8.17\ V}$

(d) $V_C = V_{CC} - I_C R_C$
$= 16\ \text{V} - (1.71\ \text{mA})(3.9\ \text{k}\Omega)$
$= \mathbf{9.33\ V}$

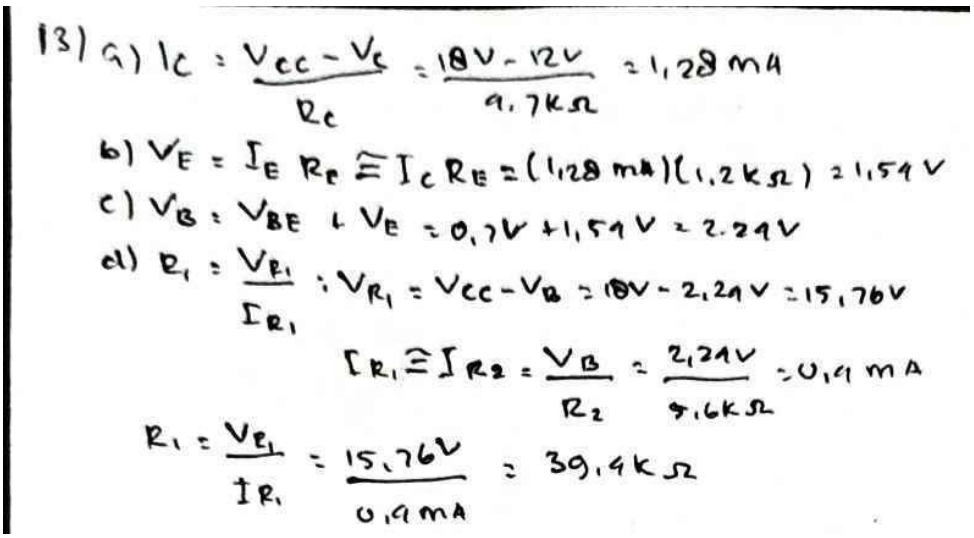
(e) $V_E = I_E R_E \cong I_C R_E = (1.71\ \text{mA})(0.68\ \text{k}\Omega)$
$= \mathbf{1.16\ V}$

(f) $V_B = V_E + V_{BE} = 1.16\ \text{V} + 0.7\ \text{V}$
$= \mathbf{1.86\ V}$

13. Given the information provided in Fig. 4.81, determine:

(a) IC. (b) VE. (c) VB. (d) R1.

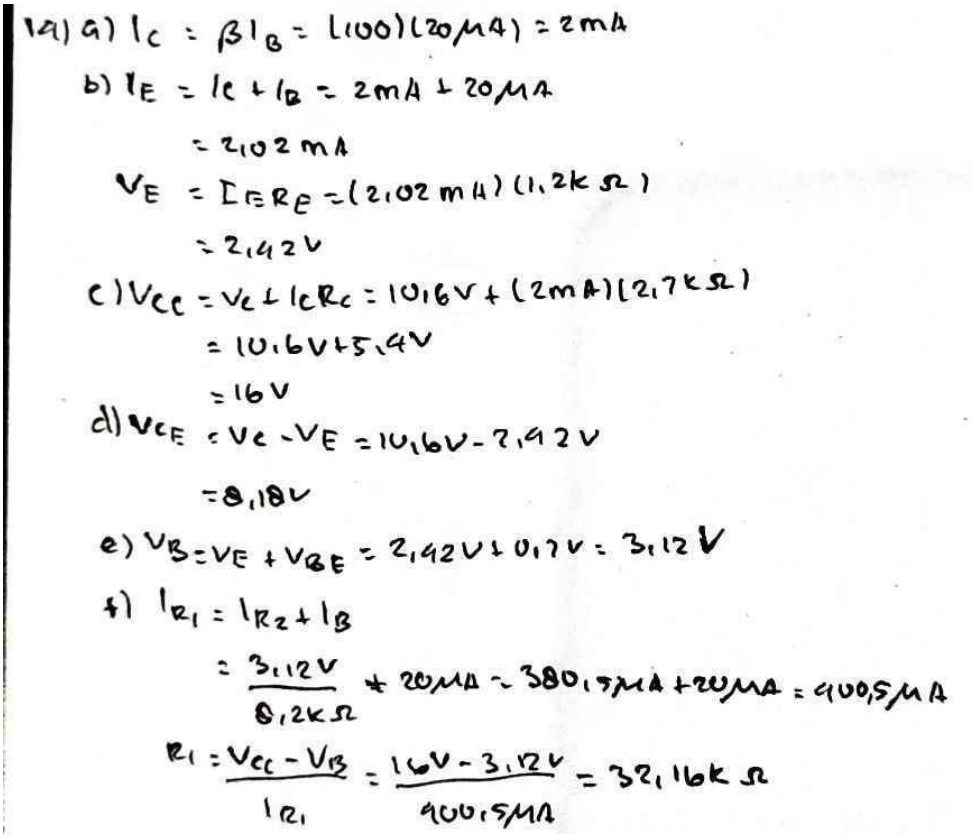
JAWAB



14. Given the information appearing in Fig. 4.82, determine:

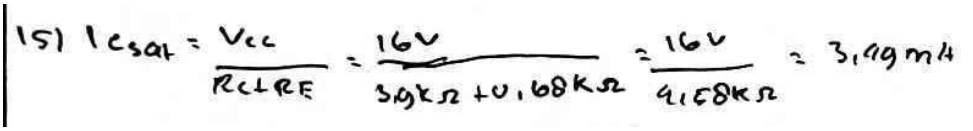
(a) IC. (b) VE. (c) VCC. (d) VCE. (e) VB. (f) R1

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15. Determine the saturation current (ICsat) for the network of Fig. 4.80.

JAWAB



16. Determine the following for the voltage-divider configuration of Fig. 4.83 using the approximate approach if the condition established by Eq. (4.33) is satisfied.

(a) IC. (b) VCE. (c) IB. (d) VE. (e) VB.

JAWAB

16) a) $\beta R_E \geq 10 R_2$

$(120)(1K\Omega) \geq 10(0,2K\Omega)$

$120K\Omega \geq 02K\Omega$

$\therefore V_B = \frac{R_2 V_{CC}}{R_1+R_2} = \frac{(0,2K\Omega)(10V)}{39K\Omega + 0,2K\Omega} = 3,13V$

$V_E = V_B - V_{BE} = 3,13V - 0,7V = 2,43V$

$I_C \cong I_E = \frac{V_E}{R_E} = \frac{2,43V}{1K\Omega} = 2,43mA$

b) $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 10V - (2,43mA)(3,3K\Omega + 1K\Omega)$

$= 7,55V$

c) $I_B = \frac{I_C}{\beta} = \frac{2,43mA}{120} = 20,25\mu A$

d) $V_E = I_E R_E \cong (2,43mA)(1K\Omega) = 2,43V$

e) $V_B = 3,13V$

17. Repeat Problem 16 using the exact (Thévenin) approach and compare solutions. Based on the results, is the approximate approach a valid analysis technique if Eq.

JAWAB

17) a) $R_{TH} = R_1 \| R_2 = 39K\Omega \| 0,2K\Omega = 0,19K\Omega$

$E_{TH} = \frac{R_2 V_{CC}}{R_1+R_2} = \frac{0,2K\Omega(10V)}{39K\Omega + 0,2K\Omega} = 3,13V$

$I_B = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E} = \frac{3,13V - 0,7V}{6,19K\Omega + (121)(1K\Omega)}$

$= \frac{2,43}{127,19K\Omega} = 19,102\mu A$

$I_C = \beta I_B = (120)(19,102\mu A) = 2,29mA$ (vs. 2,43mA)

b) $V_{CE} = V_{cc} - I_C(R_C + R_C) = 10V - (2,28\ mA)(3,3k\Omega + 1k\Omega)$
$= 10V - 9,18V = 0,82V$

c) 19,02 μA

d) $V_E = I_E R_E \cong I_C R_E = (2,28\ mA)(1k\Omega) = 2,28V$

e) $V_B = V_{BE} + V_E = 0,7V + 2,28V = 2,98V$

18. (a) Determine ICQ, VCEQ, and IBQ for the network of Problem 12 (Fig. 4.80) using the approximate approach even though the condition established by Eq. (4.33) is not satisfied.

(b) Determine ICQ, VCEQ, and IBQ using the exact approach.

(c) Compare solutions and comment on whether the difference is sufficiently large to require standing by Eq. (4.33) when determining which approach to employ.

JAWAB

18) a) $V_B = \frac{R_2}{R_1+R_E} V_{cc} = \frac{9,1k\Omega(16V)}{62k\Omega + 9,1k\Omega} = 2,05V$

$V_E = V_B - V_{BE} = 2,05V - 0,7V = 1,35V$

$I_E = \frac{V_E}{R_E} = \frac{1,35V}{0,68k\Omega} = 1,99mA$

$I_{CQ} \cong I_E = 1,99mA$

$V_{CE\,Q} = V_{cc} - I_C(R_C + R_E)$
$= 16V - (1,99mA)(3,9k\Omega + 0,68k\Omega)$
$= 16V - 9,11V$
$= 6,89V$

$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{1,99mA}{80} = 24,88\mu A$

b) prob 12:
$I_{CQ} = 1,91mA, V_{CEQ} = 8,17V, I_{BQ} = 21,92\mu A$

c) The differences of about 14% suggest that the exact approach should be employed when appropriate

19. (a) Using the characteristics of Fig. 4.76, determine RC and RE for a voltage-divider network having a Q-point of ICQ 5 mA and VCEQ 8 V. Use VCC 24 V and RC 3RE.

(b) Find VE.

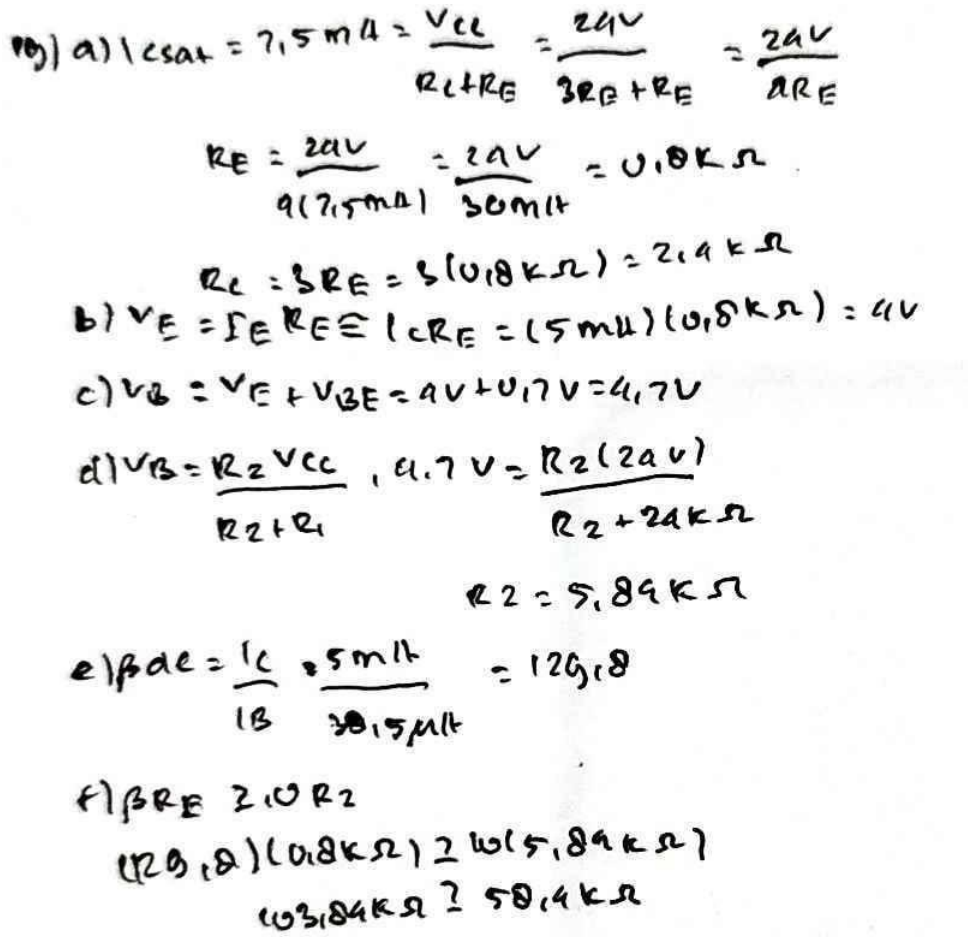
(c) Determine VB.

(d) Find R2 if R1 24 k assuming that ———————————RE 10R2.

(e) Calculate ——————————— at the Q-point.

(f) Test Eq. (4.33), and note whether the assumption of part (d) is correct.

JAWAB



20. (a) Determine IC and VCE for the network of Fig. 4.80.

(b) Change ———————————— to 120 (50% increase), and determine the new values of IC and VCE for the network of Fig. 4.80.

(c) Determine the magnitude of the percent change in IC and VCE using the following equations:

(d) Compare the solution to part (c) with the solutions obtained for parts (c) and (f) of Problem 11. If not performed, note the solutions provided in Appendix E. (e) Based on the results of part (d), which configuration is least sensitive to variations in ————————————?

JAWAB

20 a) prob 12, $I_C = 1.71$ mA

prob 12c, $V_{CE} = 8.17$ V

b) β changed to 120:

prob 12a, $E_{Th} = 2.05$ V, $R_{Th} = 7.94$ kΩ

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05\text{V} - 0.7\text{V}}{7.94\text{K}\Omega + (121)(0.68\text{K}\Omega)}$$

$= 14.96\ \mu$A

$I_C = \beta I_B = (120)(14.96\ \mu\text{A}) = 1.8$ mA

$V_{CE} = V_{CC} - I_C(R_C + R_E)$

$= 16\text{V} - (1.8\text{mA})(3.9\text{K}\Omega + 0.68\text{K}\Omega)$

$= 7.76$ V

c) $\%\Delta I_C = \left| \frac{1.8\text{mA} - 1.71\text{mA}}{1.71\text{mA}} \right| \times 100\% = 5.26\%$

$\%\Delta V_{CE} = \left| \frac{7.76\text{V} - 8.17\text{V}}{8.17\text{V}} \right| \times 100\% = 5.02\%$

d)

	I_C	I_B	V_C	
$\%\Delta I_C$	89.85%	94.79%	5.26%	voltage-divider
$\%\Delta V_{CE}$	89.20%	86.26%	5.02%	
	fixed-bias	emitter feedback		

e) Quite obviously, the voltage divider configuration is the least sensitive to changes in β.

21. a) Repeat parts (a) through (e) of Problem 20 for the network of Fig. 4.83. Change β to 180 in part (b).

(b) What general conclusions can be made about networks in which the condition β RE > 10R2 is satisfied and the quantities IC and VCE are to be determined in response to a change in β?

➢

(a) $R' = R_C + R_E = 8.2\ \text{k}\Omega + 1.8\ \text{k}\Omega = 10\ \text{k}\Omega$

$R_F/\beta = 550\ \text{k}\Omega/180 = 3.06\ \text{k}\Omega$

$R' \cong \mathbf{3.27(}R_F/\beta\mathbf{)}$ should be close

(b) $$I_{C_Q} = \frac{V_{CC} - V_{BE}}{R_C + R_E} = \frac{30\ \text{V} - 0.7\ \text{V}}{8.2\ \text{k}\Omega + 1.8\ \text{k}\Omega} = \mathbf{2.93\ mA}$$

$I_{C_Q} = \mathbf{2.93\ mA}$ relatively close to 2.24 mA

22. For the collector feedback configuration of Fig. 4.84, determine:

(a) IB.

(b) IC.

(c) VC.

➢

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{16\text{ V} - 0.7\text{ V}}{270\text{ k}\Omega + (120)(3.6\text{ k}\Omega + 1.2\text{ k}\Omega)}$$
$$= \mathbf{18.09\ \mu A}$$

(b) $$I_C = \beta I_B = (120)(18.09\ \mu\text{A})$$
$$= \mathbf{2.17\ mA}$$

(c) $$V_C = V_{CC} - I_C R_C$$
$$= 16\text{ V} - (2.17\text{ mA})(3.6\text{ k}\Omega)$$
$$= \mathbf{8.19\ V}$$

23. For the voltage feedback network of Fig. 4.85, determine:

(a) IC.

(b) VC.

(c) VE.

(d) VCE.

➢

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{30\text{ V} - 0.7\text{ V}}{550\text{ k}\Omega + 180(8.2\text{ k}\Omega + 1.8\text{ k}\Omega)} = 12.47\ \mu\text{A}$$

$I_C = \beta I_B = (180)(12.47\ \mu\text{A}) = \mathbf{2.24\ mA}$

(b) $V_C = V_{CC} - I_C R_C$
$= 30\text{ V} - (2.24\text{ mA})(8.2\text{ k}\Omega) = 30\text{ V} - 18.37\text{ V} = \mathbf{11.63\ V}$

(c) $V_E = I_E R_E \cong I_C R_E = (2.24\text{ mA})(1.8\text{ k}\Omega) = \mathbf{4.03\ V}$

(d) $V_{CE} = V_{CC} - I_C(R_C + R_E) = 30\text{ V} - (2.24\text{ mA})(8.2\text{ k}\Omega + 1.8\text{ k}\Omega)$
$= \mathbf{7.6\ V}$

24.

(a) Determine the level of I_C and V_{CE} for the network of Fig. 4.86.
(b) Change β to 135 (50% increase), and calculate the new levels of I_C and V_{CE}.
(c) Determine the magnitude of the percent change in I_C and V_{CE} using the following equations:

$$\%\Delta I_C = \left| \frac{I_{C_{(\text{part b})}} - I_{C_{(\text{part a})}}}{I_{C_{(\text{part a})}}} \right| \times 100\%, \qquad \%\Delta V_{CE} = \left| \frac{V_{CE_{(\text{part b})}} - V_{CE_{(\text{part a})}}}{V_{CE_{(\text{part a})}}} \right| \times 100\%$$

(d) Compare the results of part (c) with those of Problems 11(c), 11(f), and 20(c). How does the collector-feedback network stack up against the other configurations in sensitivity to changes in β?

➢

(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22\text{ V} - 0.7\text{ V}}{470\text{ k}\Omega + (90)(9.1\text{ k}\Omega + 9.1\text{ k}\Omega)}$$
$$= 10.09\ \mu\text{A}$$
$$I_C = \beta I_B = (90)(10.09\ \mu\text{A}) = \mathbf{0.91\text{ mA}}$$
$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22\text{ V} - (0.91\text{ mA})(9.1\text{ k}\Omega + 9.1\text{ k}\Omega)$$
$$= \mathbf{5.44\text{ V}}$$

(b) $\beta = 135$, $$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22\text{ V} - 0.7\text{ V}}{470\text{ k}\Omega + (135)(9.1\text{ k}\Omega + 9.1\text{ k}\Omega)}$$
$$= 7.28\ \mu\text{A}$$

$$I_C = \beta I_B = (135)(7.28\ \mu\text{A}) = \mathbf{0.983\text{ mA}}$$
$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22\text{ V} - (0.983\text{ mA})(9.1\text{ k}\Omega + 9.1\text{ k}\Omega)$$
$$= \mathbf{4.11\text{ V}}$$

(c) $$\%\Delta I_C = \left|\frac{0.983\text{ mA} - 0.91\text{ mA}}{0.91\text{ mA}}\right| \times 100\% = \mathbf{8.02\%}$$
$$\%\Delta V_{CE} = \left|\frac{4.11\text{ V} - 5.44\text{ V}}{5.44\text{ V}}\right| \times 100\% = \mathbf{24.45\%}$$

(d) The results for the collector feedback configuration are closer to the voltage-divider configuration than to the other two. However, the voltage-divider configuration continues to have the least sensitivities to change in β.

25. Determine the range of possible values for VC for the network of Fig. 4.87 using the 1M ohm potentiometer

➢

1 MΩ = 0 Ω, $R_B = 150\text{ k}\Omega$
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\text{ V} - 0.7\text{ V}}{150\text{ k}\Omega + (180)(4.7\text{ k}\Omega + 3.3\text{ k}\Omega)}$$
$$= 7.11\ \mu\text{A}$$
$$I_C = \beta I_B = (180)(7.11\ \mu\text{A}) = 1.28\text{ mA}$$
$$V_C = V_{CC} - I_C R_C = 12\text{ V} - (1.28\text{ mA})(4.7\text{ k}\Omega)$$
$$= \mathbf{5.98\text{ V}}$$

Full 1 MΩ: $R_B = 1{,}000\text{ k}\Omega + 150\text{ k}\Omega = 1{,}150\text{ k}\Omega = 1.15\text{ M}\Omega$
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\text{ V} - 0.7\text{ V}}{1.15\text{ M}\Omega + (180)(4.7\text{ k}\Omega + 3.3\text{ k}\Omega)}$$
$$= 4.36\ \mu\text{A}$$
$$I_C = \beta I_B = (180)(4.36\ \mu\text{A}) = 0.785\text{ mA}$$
$$V_C = V_{CC} - I_C R_C = 12\text{ V} - (0.785\text{ mA})(4.7\text{ k}\Omega)$$
$$= \mathbf{8.31\text{ V}}$$
V_C ranges from **5.98 V to 8.31 V**

26. Given VB = 4 V for the network of Fig. 4.88, determine:

(a) VE.

(b) IC.

(c) VC.

(d) VCE.

(e) IB

➢

(a) $V_E = V_B - V_{BE} = 4\text{ V} - 0.7\text{ V} = \mathbf{3.3\text{ V}}$

(b) $I_C \cong I_E = \dfrac{V_E}{R_E} = \dfrac{3.3\text{ V}}{1.2\text{ k}\Omega} = \mathbf{2.75\text{ mA}}$

(c) $V_C = V_{CC} - I_C R_C = 18\text{ V} - (2.75\text{ mA})(2.2\text{ k}\Omega)$

$= \mathbf{11.95\text{ V}}$

(d) $V_{CE} = V_C - V_E = 11.95\text{ V} - 3.3\text{ V} = \mathbf{8.65\text{ V}}$

(e) $I_B = \dfrac{V_{R_B}}{R_B} = \dfrac{V_C - V_B}{R_B} = \dfrac{11.95\text{ V} - 4\text{ V}}{330\text{ k}\Omega} = \mathbf{24.09\ \mu A}$

27. Given VC = 8 V for the network of Fig. 4.89, determine:

(a) IB.

(b) IC.

(c) β.

(d) VCE.

➢

(a) $I_B = \dfrac{V_{R_B}}{R_B} = \dfrac{V_C - V_{BE}}{R_B} = \dfrac{8\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} = \mathbf{13.04\ \mu A}$

(b) $I_C = \dfrac{V_{CC} - V_C}{R_C} = \dfrac{18\text{ V} - 8\text{ V}}{3.9\text{ k}\Omega} = \dfrac{10\text{ V}}{3.9\text{ k}\Omega} = \mathbf{2.56\text{ mA}}$

(c) $\beta = \dfrac{I_C}{I_B} = \dfrac{2.56\text{ mA}}{13.04\ \mu A} = \mathbf{196.32}$

(d) $V_{CE} = V_C = \mathbf{8\text{ V}}$

28. Untuk jaringan pada Gambar 4.90, tentukan:

(a) IB.

(b) IC.

(c) VCE.

(d) VC.

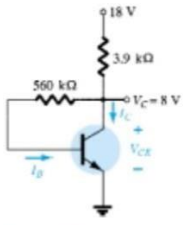


Figure 4.89 Problem 27

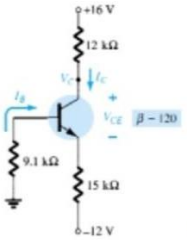


Figure 4.90 Problem 28

Jawab :

- **IB = VEE−VBE = 12V−0.7V**
  = RB +(β+1)RE 9.1kΩ+(120+1)15kΩ

  = 6.2 μA
- **IC = βIB = (120)(6.2 μA) = 0.744 mA**
- **VCE =VCC +VEE −IC(RC +RE)**
  = 16 V + 12 V − (0.744 mA)(27 kΩ) = 7.91 V
- **VC =VCC −ICRC =16V−(0.744mA)(12kΩ)=7.07V**

29. Untuk jaringan Gambar 4.91, tentukan:

(a) IB.

(b) IC.

(c) VE.

(d) VCE.

Jawab :

(a) $I_E = \dfrac{8\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} = \dfrac{7.3\text{ V}}{2.2\text{ k}\Omega} = \mathbf{3.32\text{ mA}}$

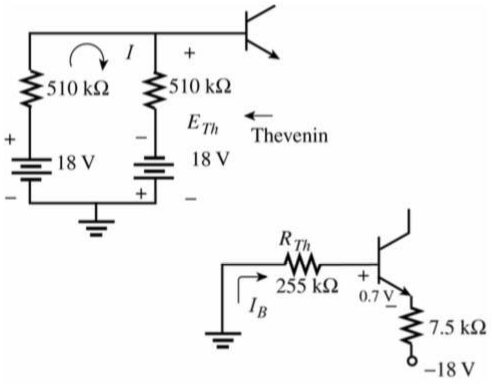
(b) $V_C = 10\text{ V} - (3.32\text{ mA})(1.8\text{ k}\Omega) = 10\text{ V} - 5.976$
$= \mathbf{4.02\text{ V}}$

(c) $V_{CE} = 10\text{ V} + 8\text{ V} - (3.32\text{ mA})(2.2\text{ k}\Omega + 1.8\text{ k}\Omega)$
$= 18\text{ V} - 13.28\text{ V}$
$= \mathbf{4.72\text{ V}}$

30. Tentukan level VE dan IE untuk jaringan pada Gambar 4.92.

Jawab :

(a) $\beta R_E > 10R_2$ not satisfied $\therefore$ Use exact approach:
Network redrawn to determine the Thevenin equivalent:



$$R_{Th} = \frac{510\ \text{k}\Omega}{2} = \mathbf{255\ k\Omega}$$

$$I = \frac{18\ \text{V} + 18\ \text{V}}{510\ \text{k}\Omega + 510\ \text{k}\Omega} = 35.29\ \mu\text{A}$$

$$E_{Th} = -18\ \text{V} + (35.29\ \mu\text{A})(510\ \text{k}\Omega) = \mathbf{0\ V}$$

$$I_B = \frac{18\ \text{V} - 0.7\ \text{V}}{255\ \text{k}\Omega + (130+1)(7.5\ \text{k}\Omega)} = \mathbf{13.95\ \mu A}$$

(b) $I_C = \beta I_B = (130)(13.95\ \mu\text{A}) = \mathbf{1.81\ mA}$

(c) $V_E = -18\ \text{V} + (1.81\ \text{mA})(7.5\ \text{k}\Omega)$
$= -18\ \text{V} + 13.58\ \text{V}$
$= \mathbf{-4.42\ V}$

(d) $V_{CE} = 18\ \text{V} + 18\ \text{V} - (1.81\ \text{mA})(9.1\ \text{k}\Omega + 7.5\ \text{k}\Omega)$
$= 36\ \text{V} - 30.05\ \text{V} = \mathbf{5.95\ V}$

31. Untuk jaringan Gambar 4.93, tentukan:

(a) IE.

(b) VC.

(c) VCE.

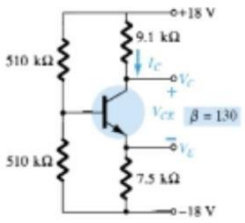


Figure 4.91 Problem 29

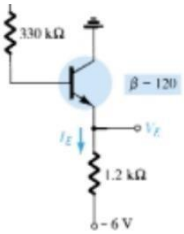


Figure 4.92 Problem 30

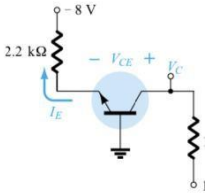


Figure 4.93 Problem 31

Jawab :

(a) $I_B = \frac{V_{R_B}}{R_B} = \frac{V_C - V_{BE}}{R_B} = \frac{8\text{ V} - 0.7\text{ V}}{560\text{ k}\Omega} = \mathbf{13.04\ \mu A}$

(b) $I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18\text{ V} - 8\text{ V}}{3.9\text{ k}\Omega} = \frac{10\text{ V}}{3.9\text{ k}\Omega} = \mathbf{2.56\ mA}$

(c) $\beta = \frac{I_C}{I_B} = \frac{2.56\text{ mA}}{13.04\ \mu\text{A}} = \mathbf{196.32}$

(d) $V_{CE} = V_C = \mathbf{8\ V}$

32. Tentukan RC dan RB untuk konfigurasi bias tetap jika VCC 12 V, 80, dan ICQ 2,5 mA dengan VCEQ 6 V. Gunakan nilai standar.

Jawab :

$$I_B = \frac{I_C}{\beta} = \frac{2.5\text{ mA}}{80} = 31.25\ \mu\text{A}$$

$$R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12\text{ V} - 0.7\text{ V}}{31.25\ \mu\text{A}} = \mathbf{361.6\ k\Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - V_{CE_Q}}{I_{C_Q}} = \frac{12\text{ V} - 6\text{ V}}{2.5\text{ mA}} = \frac{6\text{ V}}{2.5\text{ mA}}$$

$$= \mathbf{2.4\ k\Omega}$$

Standard values:

$R_B = \mathbf{360\ k\Omega}$

$R_C = \mathbf{2.4\ k\Omega}$